

POLYNOMIAL LEMNISCATES MEET IN AT MOST $2n_1n_2 - 2$ POINTS

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ABSTRACT. Orevkov and Pakovich proved that the lemniscates $|P_1| = \rho_1$ and $|P_2| = \rho_2$ of two rational functions of degrees n_1, n_2 meet in at most $2n_1n_2$ points when they meet in finitely many, that this is sharp for rational functions, and remarked that it does not seem sharp for polynomials. We show that two polynomial lemniscates of degrees $n_1, n_2 \geq 2$ with finitely many common points have at most $2n_1n_2 - 2$ of them. The bound is attained for $(n_1, n_2) = (2, 2)$ and $(2, 3)$, so the maximum is 6 and 10 there; in particular two distinct Cassini ovals, and two Bernoulli lemniscates, meet in at most six points, which answers a question from Mathematics Stack Exchange. The proof treats z and $\bar{z}$ as independent variables: the $2n_1n_2$ complex solutions, counted with multiplicity, have a weighted sum that equals $-\frac{n_1n_2}{2}|c_1 - c_2|^2 \leq 0$ for explicit centres c_1, c_2 , and this forces two of them off the real locus. Every proof is formally verified in Lean.

Status of this preprint. Every theorem, proposition, lemma and corollary is formally verified in Lean 4 with Mathlib, using only Lean’s standard axioms (Section 9), except that the ten common points in Corollary 3 are certified by interval arithmetic and rechecked by an independent program. This version has not yet been independently reviewed.

1. INTRODUCTION

How many points can two lemniscates of Bernoulli have in common? The question was asked on Mathematics Stack Exchange in 2015 [3]. A lemniscate of Bernoulli is a quartic curve with nodes at the two circular points at infinity, so Bézout’s theorem gives 16 intersections in the complex projective plane, of which at least 8 sit at the circular points; that leaves at most 8 real ones. Experiments suggested that 6 is the truth, and the asker wanted to know where the other two complex points are.

Figure 1 shows six common points. The lemniscate with foci ± 1 is $|z^2 - 1| = 1$, the one with foci $0, 2$ is $|(z - 1)^2 - 1| = 1$, and they cross at $z = \frac{1}{2} \pm \frac{i}{2}\sqrt{4\sqrt{2} - 5}$ and $z = \frac{1}{2} \pm \sqrt{2/3} \pm \frac{i}{\sqrt{12}}$. We prove that six is the most possible, and we prove it as the case $n_1 = n_2 = 2$ of a statement about all polynomial lemniscates.

For a nonconstant rational function P and $\rho > 0$, the lemniscate $\{z \in \mathbb{C} : |P(z)| = \rho\}$ is the set of real points of an algebraic curve of degree $2 \deg P$. Orevkov and Pakovich [4] proved that two such lemniscates, of degrees n_1 and n_2 , have at most $2n_1n_2$ common points when they have finitely many, and that $2n_1n_2$ is attained by rational functions of every pair of degrees; this sharpened the bound $(n_1 + n_2)^2$ of Pakovich and Shparlinski [5, Theorem 2.2]. For polynomials they constructed lemniscates with $M(n_1, n_2) = n_1n_2 + n_2 + de$ common points, where $n_1 \leq n_2$, $d = \gcd(n_1, n_2)$, and e is 1 if n_1/d is even and 0 otherwise, and they remarked that $2n_1n_2$ “does not seem” sharp for polynomials [4, Section 2.5]. For $n_1 = 1$ (a circle) the two numbers agree, $M(1, n_2) = 2n_2$. Our main result settles the remark for all other degrees.

Theorem 1. *Let $P_1, P_2 \in \mathbb{C}[z]$ have degrees $n_1, n_2 \geq 2$, let $\rho_1, \rho_2 > 0$ and $\Gamma_k = \{z \in \mathbb{C} : |P_k(z)| = \rho_k\}$. If $\Gamma_1 \cap \Gamma_2$ is finite, then*

$$\#(\Gamma_1 \cap \Gamma_2) \leq 2n_1n_2 - 2.$$

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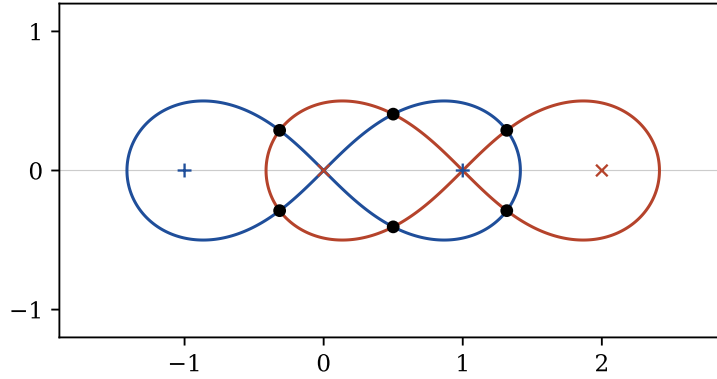


FIGURE 1. The Bernoulli lemniscates $|z^2 - 1| = 1$ (blue, foci ± 1) and $|(z-1)^2 - 1| = 1$ (red, foci $0, 2$) meet in six points (black).

Corollary 2. *Two distinct Cassini ovals $|(z-a)(z-b)| = \rho$ meet in at most six points. In particular two distinct lemniscates of Bernoulli meet in at most six points, and six is attained (Figure 1).*

Corollary 3. *The maximum number of common points of two polynomial lemniscates of degrees 2 and 3 with finitely many common points is 10, attained by $|(z - \frac{1}{10})^2 - 1| = 1$ and $|z^3/8000 - 1| = 1$. The pairs $(2, 2)$ and $(2, 3)$ are the only ones, with $2 \leq n_1 \leq n_2$, for which $M(n_1, n_2) = 2n_1n_2 - 2$.*

The idea. Write $P_k = \lambda_k f_k$ with f_k monic, so that $z \in \Gamma_k$ means $f_k(z)\overline{f_k(z)} = r_k$ with $r_k = (\rho_k/|\lambda_k|)^2$. Replace $\bar{z}$ by a second variable W and the conjugate value $\overline{f_k(z)}$ by $g_k(W)$, where g_k has the conjugate coefficients of f_k . The system

$$(1) \quad f_1(Z)g_1(W) = r_1, \quad f_2(Z)g_2(W) = r_2$$

has, when f_1 and f_2 are coprime, exactly $2n_1n_2$ solutions $(Z, W) \in \mathbb{C}^2$ counted with multiplicity, and a point z lies on both lemniscates exactly when $(z, \bar{z})$ is a solution. The map $\sigma(Z, W) = (\bar{W}, Z)$ permutes the solutions, and its fixed points are the pairs $(z, \bar{z})$. The heart of the proof is the identity

$$(2) \quad \sum_{\text{solutions}} (Z - m)(W - \bar{m}) = -\frac{n_1n_2}{2} |c_1 - c_2|^2, \quad m = \frac{c_1 + c_2}{2},$$

where c_k is the centroid of the roots of f_k . In words: after centring at the midpoint of the two centroids, the weighted sum of the products $(Z - m)(W - \bar{m})$ over all solutions is a real number that is never positive. A real solution $(z, \bar{z})$ contributes $|z - m|^2 \geq 0$. So if all solutions were real, every one of them would sit at $z = m$; hence at least one solution is not real, and with it its partner under σ . That accounts for two of the $2n_1n_2$ solutions and gives Theorem 1 in the coprime case (Section 2). For the two Bernoulli lemniscates of Figure 1, $c_1 = 0$, $c_2 = 1$, the sum is -2 , and the two missing points the asker looked for are the solutions $(\frac{1}{2} \mp t, \frac{1}{2} \pm t)$ with $t^2 = \frac{5}{4} + \sqrt{2}$ (Figure 2).

Identity (2) is proved by deformation. The left side is a trace (Section 3); as the level r_1 varies it is a polynomial in r_1 (Section 4); the solutions grow at most like $|r_1|^{1/2}$, so the polynomial is constant (Section 5); and at level 0 the algebra splits into two tensor products whose traces are computed directly (Section 6). A second deformation, in r_2 , brings the second level to 0 as well. No intersection theory is needed: the solution count $2n_1n_2$ comes out of the same computation at level 0. When f_1 and f_2 share a root (Section 7) we eliminate W : the resultant $R(Z)$ vanishes at every common point, and a shared root removes its top two coefficients, so it has at most $2n_1n_2 - 2$ roots unless it vanishes identically, in which case the two lemniscates turn out to be nested.

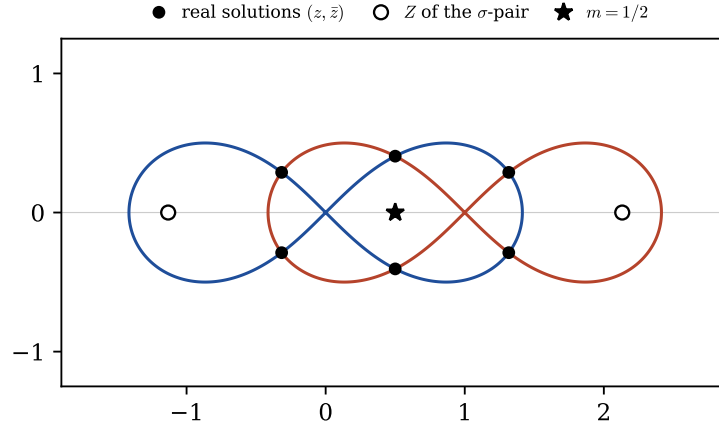


FIGURE 2. The eight solutions of (1) for the lemniscates of Figure 1, drawn by their Z -coordinate. Six are real, $W = \bar{Z}$, and contribute $|Z - \frac{1}{2}|^2$, in total $\frac{1}{2} + 2\sqrt{2}$. The other two are $(\frac{1}{2} - t, \frac{1}{2} + t)$ and its image $(\frac{1}{2} + t, \frac{1}{2} - t)$ under σ , with $t^2 = \frac{5}{4} + \sqrt{2}$; each contributes $(Z - \frac{1}{2})(W - \frac{1}{2}) = -(\frac{5}{4} + \sqrt{2})$. The total is -2 , as (2) predicts.

2. DOUBLING THE VARIABLE

For $f \in \mathbb{C}[z]$ write $\bar{f}$ for the polynomial with conjugate coefficients, so that $\bar{f}(\bar{z}) = \overline{f(z)}$. Throughout, f_1, f_2 are monic of degrees $n_1, n_2 \geq 2$, $r_1, r_2 > 0$, and $g_k = \bar{f}_k$. Put

$$F_k(Z, W) = f_k(Z)g_k(W) - r_k \in \mathbb{C}[Z, W], \quad \sigma(Z, W) = (\bar{W}, \bar{Z}).$$

Then $F_k(z, \bar{z}) = |f_k(z)|^2 - r_k$, so $z \in \Gamma_k$ if and only if $F_k(z, \bar{z}) = 0$, and $F_k(\sigma(Z, W)) = f_k(\bar{W})g_k(\bar{Z}) - r_k = \overline{F_k(Z, W)}$, so σ maps the common zero set of F_1, F_2 to itself. The fixed points of σ are exactly the pairs $(z, \bar{z})$.

The next lemma is the counting step. It says that a multiset of solutions closed under σ , with a nonpositive weighted sum, always leaves two of its members off the real locus.

Lemma 4. *Let S be a finite multiset of points of $\mathbb{C}^2$ with $N \geq 3$ members, such that $\sigma(p)$ is a member of S whenever p is, and let $m \in \mathbb{C}$ with $\sum_{p \in S} (Z_p - m)(W_p - \bar{m})$ a real number ≤ 0 . Then at most $N - 2$ distinct points of the form $(z, \bar{z})$ are members of S .*

Proof. Suppose $N - 1$ distinct points $(z, \bar{z})$ are members. If some member q is not of this form, then $\sigma(q) \neq q$ is a member as well, also not of this form, and $q, \sigma(q)$ and the $N - 1$ fixed points give $N + 1$ members: impossible. So every member has the form $(z, \bar{z})$ and contributes $(z - m)(\bar{z} - \bar{m}) = |z - m|^2 \geq 0$. The sum is then ≥ 0 , hence 0, so every member equals $(m, \bar{m})$, contradicting the $N - 1 \geq 2$ distinct members. $\square$

3. TRACES IN A FINITE-DIMENSIONAL ALGEBRA

To speak of the solutions of (1) with multiplicity we use the quotient algebra. For a finite-dimensional commutative $\mathbb{C}$ -algebra A and $x \in A$ write $\text{Tr}_A(x)$ for the trace of multiplication by x .

Lemma 5. *Let A be a finite-dimensional commutative $\mathbb{C}$ -algebra and $a, b \in A$. There is a multiset S of points of $\mathbb{C}^2$ with $\dim A$ members such that*

- (1) $\text{Tr}_A(P(a, b)) = \sum_{p \in S} P(p)$ for every $P \in \mathbb{C}[Z, W]$;
- (2) if $P(a, b) = 0$ then $P(p) = 0$ for every member p ;

(3) $(\chi(a), \chi(b))$ is a member of S for every $\mathbb{C}$ -algebra map $\chi: A \rightarrow \mathbb{C}$.

Proof. Multiplication by a and by b commute, so A is the direct sum of the joint generalised eigenspaces $V_\chi = \{v : (a - \chi_1)^k v = (b - \chi_2)^k v = 0 \text{ for some } k\}$, $\chi \in \mathbb{C}^2$. Let S contain each χ with multiplicity $\dim V_\chi$. Let I_χ be the radical of the annihilator of V_χ , an ideal of A ; it contains $a - \chi_1$ and $b - \chi_2$, so $P(a, b) - P(\chi) \in I_\chi$ for every P . Hence $P(a, b)$ acts on V_χ as $P(\chi)$ plus a nilpotent map, whose trace is 0, and summing over χ gives (1); $P = 1$ gives $\#S = \dim A$. If $P(a, b) = 0$ then $P(\chi)^k$ annihilates $V_\chi \neq 0$, so $P(\chi) = 0$: this is (2). For (3), χ vanishes on V_ψ whenever $\psi \neq (\chi(a), \chi(b))$: if $\psi_1 \neq \chi(a)$, then $(\chi(a) - \psi_1)^k \chi(v) = \chi((a - \psi_1)^k v) = 0$ for $v \in V_\psi$, and if $\psi_2 \neq \chi(b)$ the same holds with b . Since $\chi(1) = 1$ and 1 is a sum of elements of the V_ψ , the space $V_{(\chi(a), \chi(b))}$ is nonzero. $\square$

4. DEFORMING THE LEVEL

We now let the first level vary. Fix $G = f'(Z)g'(W) - r$ and $h = f(Z)g(W)$, and study the algebras $\mathbb{C}[Z, W]/(G, h - c)$ for all $c \in \mathbb{C}$ at once, as the fibres of one module over $\mathbb{C}[t]$.

Lemma 6. *Let f_1, f_2 be coprime and nonconstant, and g_1, g_2 monic, coprime and nonconstant. For every $c \in \mathbb{C}$ and $r \neq 0$, the polynomials $f_1 g_1 - c$ and $f_2 g_2 - r$ have no common prime factor in $\mathbb{C}[Z, W]$; neither do $f_1 g_1$ and $f_2 g_2 - c$.*

Proof. Let H be a prime common factor. Suppose first that H does not involve W . Then $H = Z - \zeta$ up to a unit, and substituting $Z = \zeta$ in the second polynomial gives $f_2(\zeta)g_2(W) = r$ (respectively $= c$) identically in W . As g_2 is nonconstant, $f_2(\zeta) = 0$; for the first pair this contradicts $r \neq 0$, and for the second pair the first polynomial gives $f_1(\zeta)g_1(W) = 0$ identically, so $f_1(\zeta) = 0$, contradicting the coprimality of f_1, f_2 .

So H has positive degree in W . Since g_1, g_2 are monic, the leading coefficients in W of the two polynomials are $f_1(Z)$ and $f_2(Z)$, and the leading coefficient of H divides both, so it is a constant. For the first pair with $c \neq 0$, take a root ζ of f_1 : then $H(\zeta, W)$ still has positive degree and divides $f_1(\zeta)g_1(W) - c = -c \neq 0$, which is impossible. For the first pair with $c = 0$, and for the second pair, H divides $f_1(Z)g_1(W)$, so it divides $g_1(W)$ and is $W - \omega$ up to a unit, with $g_1(\omega) = 0$. Substituting $W = \omega$ in the other polynomial gives $f_2(Z)g_2(\omega) = r$ or $= c$ identically in Z , so $g_2(\omega) = 0$ (and $r = 0$ or $c = 0$); a common root of g_1, g_2 contradicts their coprimality, and $r = 0$ contradicts $r \neq 0$. $\square$

The next lemma reads off the leading terms of a resultant from the leading terms of the entries. For polynomials P, Q in W and integers $m \geq \deg_W P$, $n \geq \deg_W Q$, let $\text{Res}^{m,n}_W(P, Q)$ be the determinant of the Sylvester matrix of size $m + n$, with m columns of shifted coefficients of Q and n of P .

Lemma 7. *Let K be a commutative ring and $P, Q \in K[Z][W]$ with $\deg_W P \leq m$, $\deg_W Q \leq n$, every W -coefficient of P of Z -degree at most d_1 and of Q at most d_2 . Let $\hat{P}, \hat{Q} \in K[W]$ collect the coefficients of Z^{d_1} in the coefficients of P , and of Z^{d_2} in those of Q . Then $\text{Res}^{m,n}_W(P, Q) \in K[Z]$ has degree at most $D = md_2 + nd_1$, and its coefficient of Z^D is $\text{Res}^{m,n}(\hat{P}, \hat{Q})$. If moreover $d_1, d_2 \geq 1$ and the coefficients of Z^{d_k-1} are e_k times those of Z^{d_k} , then the coefficient of Z^{D-1} is $(me_2 + ne_1) \text{Res}^{m,n}(\hat{P}, \hat{Q})$.*

Proof. In the Leibniz expansion of the Sylvester determinant each product takes one entry from each column; the entries of a Q -column have degree $\leq d_2$, those of a P -column $\leq d_1$. A product of polynomials of degrees $\leq \delta_j$ has degree $\leq \sum \delta_j$, with top coefficient the product of the top coefficients, and its coefficient of degree $\sum \delta_j - 1$ is $\sum_j (\text{coefficient of degree } \delta_j - 1 \text{ in factor } j) \prod_{i \neq j} (\text{top of factor } i)$. Under the proportionality assumption this is $(\sum_j e_j) \prod_j (\text{top})$. Summing over permutations gives the two claims, since the matrix of top coefficients is the Sylvester matrix of $\hat{P}, \hat{Q}$. $\square$

Proposition 8. *Let f, g, f', g' be monic of positive degree, f, f' coprime, g, g' coprime, $r \in \mathbb{C}$, and assume that for no c do $h - c$ and G have a common prime factor, where $h = f(Z)g(W)$ and $G = f'(Z)g'(W) - r$. Let $M = \mathbb{C}[Z, W]/(G)$, a $\mathbb{C}[t]$ -algebra by $t \mapsto h$. Then M is a free $\mathbb{C}[t]$ -module of finite rank N . For every $c \in \mathbb{C}$ the fibre $Q_c = M/(t - c)M = \mathbb{C}[Z, W]/(G, h - c)$ has dimension N , and for every $P \in \mathbb{C}[Z, W]$ there is a polynomial $\tau_P \in \mathbb{C}[t]$, not depending on c , with $\text{Tr}_{Q_c}(P) = \tau_P(c)$.*

Proof. Finite. In $\mathbb{C}[t][Z][W]$ the resultant $\text{Res}_W(f(Z)g(W) - t, G)$ is a combination of the two polynomials, so it vanishes in M when $t = h$. Its W -coefficients are $f(Z) \cdot (\text{coefficient of } g) \text{ minus } t$ in degree 0, and $f'(Z) \cdot (\text{coefficient of } g') \text{ minus } r$; with $d_1 = \deg f \geq 1$, $d_2 = \deg f'$, Lemma 7 shows that it has Z -degree $\deg g \deg f' + \deg g' \deg f$ with top coefficient $\text{Res}(g, g')$, a nonzero constant because g, g' are coprime. Dividing by it, Z is a root of a monic polynomial over $\mathbb{C}[t]$; exchanging the roles of Z and W (using that f, f' are coprime) does the same for W . As M is generated by Z and W , it is a finite $\mathbb{C}[t]$ -module.

Torsion-free. Let $p \neq 0$ and $p(h)a \in (G)$. Factor $p = \lambda \prod_j (t - c_j)$. A prime factor of G that divided $p(h) = \lambda \prod_j (h - c_j)$ would divide some $h - c_j$, which is excluded, so G divides a .

Free. A finite torsion-free module over the principal ideal domain $\mathbb{C}[t]$ is free.

Fibres. Let $b_1, \dots, b_N$ be a $\mathbb{C}[t]$ -basis of M and write $s = \sum_i \beta_i(s)b_i$ with $\beta_i(s) \in \mathbb{C}[t]$. The map $s \mapsto (\beta_i(s)(c))_i$ vanishes on $(t - c)M$, and $s - \sum_i \beta_i(s)(c)b_i = \sum_i (\beta_i(s) - \beta_i(s)(c))b_i$ lies in $(t - c)M$ because $t - c$ divides $\beta - \beta(c)$. So the images of the b_i form a $\mathbb{C}$ -basis of Q_c , and the matrix of multiplication by P on Q_c in this basis is the matrix on M with t replaced by c . Hence $\text{Tr}_{Q_c}(P) = \tau_P(c)$, where τ_P is the trace over $\mathbb{C}[t]$ of multiplication by P on M . $\square$

Combining Proposition 8 with Lemma 5, applied to $A = Q_c$, $a = Z$, $b = W$: for each c there is a multiset S_c of N common zeros of $h - c$ and G , containing each common zero at least once, with $\sum_{p \in S_c} P(p) = \tau_P(c)$. The common zeros give algebra maps $Q_c \rightarrow \mathbb{C}$, hence members, by Lemma 5(3).

5. GROWTH

A polynomial τ in one variable with $|\tau(c)| \leq A + B|c|^{1/2}$ for all c is constant, since a polynomial of degree $d \geq 1$ satisfies $|\tau(c)| \geq \frac{1}{2}|\text{lc}(\tau)| |c|^d$ for large $|c|$. So $\tau_P(c)$ is constant as soon as the members of S_c grow at most like $|c|^{1/2}$, which is what the next lemma provides.

Lemma 9. *Let $\varphi = (Z - m)(W - m')$ with $m, m' \in \mathbb{C}$.*

- (1) *Let f, g, f', g' be monic with $\deg f, \deg g \geq 2$ and $\deg f', \deg g' \geq 1$, with f, f' coprime and g, g' coprime, and $r \in \mathbb{C}$. There are A, B such that every common zero of $f(Z)g(W) - c$ and $f'(Z)g'(W) - r$ satisfies $|\varphi| \leq A + B|c|^{1/2}$, for every c .*
- (2) *Let f_1, g_1, f_2, g_2 be monic, with $\deg f_1 \geq 1$, $\deg g_1 \geq 1$, $\deg f_2 \geq 2$ and $\deg g_2 \geq 2$, and with f_1, f_2 coprime and g_1, g_2 coprime. There are A, B such that every common zero of $f_1(Z)g_1(W)$ and $f_2(Z)g_2(W) - d$ satisfies $|\varphi| \leq A + B|d|^{1/2}$, for every d .*

Proof. Two facts are used. A monic polynomial u of degree $n \geq 1$ satisfies $\frac{1}{2}|w|^n \leq |u(w)| \leq 2|w|^n$ once $|w|$ exceeds a constant. Coprime p, q are not simultaneously small on a disc: from $up + vq = 1$ and bounds for $|u|, |v|$ on the disc, $\max(|p|, |q|) \geq \delta > 0$ there.

The main step is a one-sided bound for (1): if $|W| \leq R_0$ then $|Z| \leq A' + B'|c|^{1/2}$. Indeed, for $|Z|$ large, $|f'(Z)| \geq |Z|/2$, so $|g'(W)| = |r|/|f'(Z)| < \delta$, where δ is the constant for g', g on the disc $|W| \leq R_0$; hence $|g(W)| \geq \delta$, and $|c| = |f(Z)| |g(W)| \geq \frac{1}{2}|Z|^2 \delta$ because $\deg f \geq 2$. The roles of Z, W are symmetric, and $|Z|, |W|$ cannot both be large, since then $|f'(Z)g'(W)| > |r|$. So one coordinate is bounded and the other is $O(1 + |c|^{1/2})$, which bounds $|\varphi| \leq (|Z| + |m|)(|W| + |m'|)$.

For (2), $f_1(Z)g_1(W) = 0$ means that Z is a root of f_1 or W is a root of g_1 , so one coordinate is bounded, and the one-sided bound applies with $(f, g, f', g', r) = (f_2, g_2, f_1, g_1, 0)$ or its mirror image. $\square$

6. LEVEL ZERO AND THE TRACE IDENTITY

At level 0 the algebra splits. Write s_p for the sum of the roots of a monic polynomial p (with multiplicity), so $s_p = -(\text{coefficient of } z^{\deg p - 1})$.

Lemma 10. *Let f, f', g, g' be monic of positive degree with f, f' coprime and g, g' coprime. Then*

$$\mathbb{C}[Z, W]/(f(Z)g(W), f'(Z)g'(W)) \cong \mathbb{C}[Z, W]/(f', g) \times \mathbb{C}[Z, W]/(f, g')$$

and $\mathbb{C}[Z, W]/(f, g) \cong \mathbb{C}[Z]/(f) \otimes \mathbb{C}[W]/(g)$. Hence the algebra has dimension $\deg f' \deg g + \deg f \deg g'$, and for $\varphi = (Z - m)(W - m')$

$$\text{Tr}(\varphi) = (s_{f'} - m \deg f')(s_g - m' \deg g) + (s_f - m \deg f)(s_{g'} - m' \deg g').$$

In words: each of the two factors contributes the product of its two centred root sums.

Proof. The map to the product is surjective: if $uf + vf' = 1$, the element $u(Z)f(Z)$ is 1 modulo (f', g) and 0 modulo (f, g') . It is injective because $(f', g) \cap (f, g') = (f', g)(f, g')$ (the two ideals are comaximal: f and f' generate the unit ideal) and the four products lie in $(fg, f'g')$: $f'g'$ and fg are the generators, and, with $\tilde{u}g + \tilde{v}g' = 1$,

$$ff' = (\tilde{u}f')fg + (\tilde{v}f)f'g', \quad gg' = (ug')fg + (vg)f'g'.$$

The tensor isomorphism sends Z, W to $Z \otimes 1, 1 \otimes W$. Dimensions add over products and multiply over tensor products. So do traces, and φ is the pure tensor $(Z - m) \otimes (W - m')$ in each factor. Finally, the trace of Z on $\mathbb{C}[Z]/(f)$ is s_f (its characteristic polynomial is f), so the trace of $Z - m$ is $s_f - m \deg f$. $\square$

Proposition 11. *Let f_1, f_2 be coprime and g_1, g_2 coprime, all monic of degree at least 2, and $r_2 \neq 0$. The solutions of $f_1(Z)g_1(W) = r_1$, $f_2(Z)g_2(W) = r_2$ are the members of a multiset S with $\deg f_1 \deg g_2 + \deg f_2 \deg g_1$ members, each solution occurring at least once, and*

$$\sum_{p \in S} (Z_p - m)(W_p - m') = (s_{f_1} - m \deg f_1)(s_{g_2} - m' \deg g_2) + (s_{f_2} - m \deg f_2)(s_{g_1} - m' \deg g_1).$$

That is, the sum over the solutions at levels (r_1, r_2) equals the trace of Lemma 10 at levels $(0, 0)$.

Proof. Apply Proposition 8 to $h = f_1g_1$, $G = f_2g_2 - r_2$ (Lemma 6 gives the hypothesis), and let $S = S_{r_1}$, with sum $\tau_1(r_1)$. By Lemma 9(1), $|\tau_1(c)| \leq N(A + B|c|^{1/2})$, so τ_1 is constant and $\tau_1(r_1) = \tau_1(0)$, the trace on $\mathbb{C}[Z, W]/(f_2g_2 - r_2, f_1g_1)$. This algebra is also the fibre at r_2 of the second deformation $h = f_2g_2$, $G = f_1g_1$ (Lemma 6, second part), so the trace equals $\tau_2(r_2)$, which by Lemma 9(2) equals $\tau_2(0)$, the trace at level $(0, 0)$, given by Lemma 10. The number of members follows the same chain: both ranks equal the dimension of the common fibre, and the second rank is the dimension at level $(0, 0)$. $\square$

Proof of Theorem 1 when f_1, f_2 are coprime. Take $g_k = \overline{f_k}$; then $\deg g_k = n_k$, $s_{g_k} = \overline{s_{f_k}}$, and g_1, g_2 are coprime. Let $c_k = s_{f_k}/n_k$ be the centroid of the roots of f_k , $m = \frac{1}{2}(c_1 + c_2)$ and $m' = \overline{m}$. Then $s_{f_1} - mn_1 = \frac{n_1}{2}(c_1 - c_2)$ and $s_{f_2} - mn_2 = -\frac{n_2}{2}(c_1 - c_2)$, and Proposition 11 gives a multiset S of $N = 2n_1n_2$ solutions with

$$\sum_{p \in S} (Z_p - m)(W_p - \overline{m}) = \frac{n_1n_2}{4}(c_1 - c_2)\overline{(c_2 - c_1)} + \frac{n_1n_2}{4}(c_2 - c_1)\overline{(c_1 - c_2)} = -\frac{n_1n_2}{2}|c_1 - c_2|^2 \leq 0.$$

The two terms are equal, each being $-\frac{n_1n_2}{4}|c_1 - c_2|^2$, because $(c_1 - c_2)\overline{(c_2 - c_1)} = -|c_1 - c_2|^2$. Since σ maps solutions to solutions and every solution is a member of S , Lemma 4 applies and at

most $2n_1n_2 - 2$ points $(z, \bar{z})$ are solutions. These are the common points of the lemniscates. No finiteness assumption was used in this case. $\square$

7. SHARED ROOTS

Suppose f_1 and f_2 have a common root ζ . We eliminate W : let $R(Z) = \text{Res}_W^{n_1, n_2}(F_1, F_2)$.

Lemma 12. (1) *If $F_1(z, w) = F_2(z, w) = 0$ then $R(z) = 0$. In particular every common point z of the lemniscates is a root of R .*

(2) $\deg R \leq 2n_1n_2 - 2$.

Proof. (1) The resultant commutes with substituting $Z = z$, and $R(z) = pF_1(z, \cdot) + qF_2(z, \cdot)$ for polynomials p, q in W ; evaluate at $W = w$. (2) The W -coefficients of F_k are $f_k(Z)$ times the coefficients of g_k , minus r_k in degree 0. Their Z^{n_k} -coefficients are the coefficients of g_k and, since $n_k \geq 2$, their Z^{n_k-1} -coefficients are e_k times those, e_k being the subleading coefficient of f_k . By Lemma 7, the coefficients of $Z^{2n_1n_2}$ and $Z^{2n_1n_2-1}$ in R are multiples of $\text{Res}(g_1, g_2)$, and $\deg R \leq 2n_1n_2$. Since $g_1(\bar{\zeta}) = g_2(\bar{\zeta}) = 0$, part (1) applied to g_1, g_2 (a Bézout relation evaluated at a common root) gives $\text{Res}(g_1, g_2) = 0$. $\square$

If $R \neq 0$, distinct common points are distinct roots of R , and Theorem 1 follows from Lemma 12. It remains to see what $R \equiv 0$ means.

Lemma 13. *If $R \equiv 0$ then $f_1^{n_2} = f_2^{n_1}$.*

Proof. For Z with $f_1(Z)f_2(Z) \neq 0$, the polynomials $F_1(Z, \cdot)$ and $F_2(Z, \cdot)$ have exact degrees n_1, n_2 and resultant $R(Z) = 0$, so they have a common root w . Let ζ be a root of f_1f_2 , of multiplicity a in f_1 and b in f_2 . As $Z \rightarrow \zeta$, one of $|g_1(w)| = r_1/|f_1(Z)|$ and $|g_2(w)| = r_2/|f_2(Z)|$ tends to infinity, so $|w| \rightarrow \infty$, and the two-sided bounds $\frac{1}{2}|w|^n \leq |g_k(w)| \leq 2|w|^n$ give $|g_1(w)|^{n_2} \asymp |w|^{n_1n_2} \asymp |g_2(w)|^{n_1}$ with constants independent of Z . Thus $|f_2(Z)|^{n_1} \asymp |f_1(Z)|^{n_2}$ near ζ . Along $Z = \zeta + t$, $t \rightarrow 0^+$, the left side is $\asymp t^{bn_1}$ and the right side $\asymp t^{an_2}$, so $an_2 = bn_1$. Hence the monic polynomials $f_1^{n_2}$ and $f_2^{n_1}$ have the same roots with the same multiplicities, and are equal. $\square$

Compare the exceptional configurations in [5, Theorem 2.2] and [4, Theorem 1.1]: there too the obstruction to a finite count is that the two functions are built from a common one.

Proof of Theorem 1 when f_1, f_2 share a root. If $R \neq 0$ we are done. If $R \equiv 0$, Lemma 13 gives $|f_2|^{2n_1} = |f_1|^{2n_2}$ on $\mathbb{C}$. If the lemniscates had a common point z_0 , then $r_2^{n_1} = r_1^{n_2}$, and every $z \in \Gamma_1$ would satisfy $|f_2(z)|^{2n_1} = r_1^{n_2} = r_2^{n_1}$, that is $z \in \Gamma_2$. But Γ_1 is infinite: at a root ζ of f_1 and for each small real x , $|f_1|^2$ is below r_1 at $\zeta + x$ and above it at $\zeta + x + iY$ for large Y , so by the intermediate value theorem Γ_1 has a point with real part $\text{Re } \zeta + x$. This contradicts finiteness, so $\Gamma_1 \cap \Gamma_2$ is empty. $\square$

8. DEGREE TWO AND SHARPNESS

Proof of Corollary 2. A Cassini oval is $|f(z)| = \rho$ with $f = (z - a)(z - b)$ monic of degree 2. In the coprime case and in the case $R \neq 0$ the proof above gives at most 6 points without assuming finiteness. If $R \equiv 0$ then $f_1^2 = f_2^2$, so $(f_1 - f_2)(f_1 + f_2) = 0$ and $f_1 = f_2$, as $f_1 + f_2$ has leading coefficient 2; two ovals with the same foci and a common point have the same level, so they are equal. The six points of Figure 1 lie on both curves: substituting them into $(x^2 + y^2)^2 = 2(x^2 - y^2)$ and its translate by 1 is an identity in $\sqrt{2}$. $\square$

Proof of Corollary 3. For $f_1 = (z - \frac{1}{10})^2 - 1$ and $f_2 = z^3/8000 - 1$, interval arithmetic certifies ten pairwise disjoint boxes, each containing exactly one common point of $|f_1| = 1$ and $|f_2| = 1$ (Section 9); Theorem 1 gives at most $2 \cdot 2 \cdot 3 - 2 = 10$. This is the construction of [4, Proposition 2] with explicit parameters. For the second claim, $M(n_1, n_2) = 2n_1n_2 - 2$ means $n_2(n_1 - 1) = de + 2 \leq$

$n_1 + 2$. As $n_2 \geq n_1$, this forces $n_1(n_1 - 1) \leq n_1 + 2$, so $n_1 = 2$; then $n_2 = de + 2$, which is 3 for odd n_2 ($d = 1, e = 1$) and 2 for even n_2 ($d = 2, e = 0$). $\square$

Remark 14. For other degrees the maximum lies between $M(n_1, n_2)$ and $2n_1n_2 - 2$, and we do not know where. The smallest open cases are $(2, 4)$, with $M(2, 4) = 12$ against our bound 14, and $(3, 3)$, with $M(3, 3) = 12$ against 16.

9. FORMAL VERIFICATION AND DATA

Every statement above, including Corollaries 2 and 3 except the numerical certificate for the ten points, is proved in Lean 4 with Mathlib [1, 2], in about 3,100 lines that depend only on Lean's three standard axioms (propositional extensionality, choice and quotient soundness). The formal proof follows the text, with one difference: the constancy of the trace polynomial in Section 5 is derived from Liouville's theorem rather than from the degree argument given there. The other steps are formalised as written: joint generalised eigenspaces for Lemma 5; the Sylvester expansion for Lemma 7; an explicit basis of the fibres for Proposition 8; the Chinese remainder theorem and tensor products for Lemma 10; and Lemmas 12 and 13 for shared roots. The main statement reads

```
theorem theoremA (P1 P2 :  $\mathbb{C}[X]$ ) (hn1 :  $2 \leq P_1.\text{natDegree}$ ) (hn2 :  $2 \leq P_2.\text{natDegree}$ )
  { $\rho_1 \rho_2$  :  $\mathbb{R}$ } (h $\rho_1$  :  $0 < \rho_1$ ) (h $\rho_2$  :  $0 < \rho_2$ )
  (hfin : {z :  $\mathbb{C}$  |  $\|P_1.\text{eval } z\| = \rho_1 \wedge \|P_2.\text{eval } z\| = \rho_2$ }.Finite) :
  hfin.toFinset.card  $\leq 2 * P_1.\text{natDegree} * P_2.\text{natDegree} - 2$ 
```

The corollaries are `oval_le_six`, `bernoulli_six` and `M_eq_iff`. A mutation test plants 19 wrong variants (bounds off by one, wrong signs, dropped hypotheses) and checks that each is rejected. The ten points of Corollary 3 and the six of Corollary 2 are certified by the Krawczyk test in interval arithmetic; an independent program recomputes all $2n_1n_2$ complex solutions of both examples from an exact resultant and confirms the counts, identity (2) and the numbers in Figure 2. A symbolic computation confirms (2) for $n_1 = n_2 = 2$ with all coefficients and levels as free variables. The Lean sources, these programs and their outputs are in the folder `polynomial-lemniscates` of <https://github.com/jvvk/mathematics>.

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